

Agentic AI for Payment Integrity: Why the Auditable Agent Wins

Abdel-Basit Jimoh | Cotiviti Intern Assessment, Topic 2 | June 2026

Defining the concept

Clinical decision-making and pattern recognition in healthcare cover a set of tasks machine learning has handled separately for years. Classification assigns an input to one of a fixed set of categories, which in payment could mean sorting a claim as clean or improper. Prediction estimates an unknown value, usually a number, like what a claim should cost or a patient's readmission risk. Clustering groups similar data points together without labeled examples, so it could surface providers who bill alike and flag the ones who do not fit. Time-series anomaly detection watches data ordered over time and flags points that break from the expected pattern, like a provider's billing for one code that suddenly spikes. None of these are unique to healthcare, but each maps onto a concrete payment-integrity task. What is new is the layer that now sits on top of these methods. Agentic generative AI gives a large language model a goal, a set of tools it can call, and a loop in which it decides which tool to use, reads the result, and reasons toward a decision. Chain-of-thought prompting is the mechanism underneath that reasoning: the model works a problem in explicit steps instead of jumping to an answer, which improves accuracy on multi-step tasks (Wei et al., 2022). The ReAct framework pairs that reasoning with tool use, and the combination reduces hallucination because the model checks itself against real data instead of its own memory (Yao et al., 2023). For Cotiviti, treatment, payment, and operations (TPO) is the frame, and payment is where these methods have the clearest return.

Trends

Three things are happening at once. First, the field has shifted from single-shot generative models to agents that take multi-step action under human oversight, and payers are among the earliest adopters for claims and prior-authorization work. Second, retrieval-augmented generation (RAG) has become the standard way to keep a model honest. By retrieving source documents before it answers, a RAG system grounds its output in real policy text and cuts hallucination, which is why most recent healthcare LLM work builds on it (Amugongo et al., 2025). Third, governance is moving from principle to operations. The NIST AI Risk Management Framework already puts a GOVERN function above everything else and treats human oversight and auditability as design requirements (NIST, 2023). Newer work names the operational problem that appears once agents reach production: agent sprawl, where duplicated agents and lingering tool permissions blur accountability, answered with identity registries, kill-switch enforcement, and audit logging tied to each agent's lifecycle (Prakash et al., 2026). For a payer, that shift is not a constraint on the technology. It is what makes the technology deployable.

Opportunities

The scale of the problem is the opportunity. CMS estimated roughly \$95.5 billion in improper payments across Medicare and Medicaid in fiscal year 2025, including \$28.83 billion in Medicare fee-for-service alone (CMS, 2026). Most of that is not fraud. It is documentation and coding error, the kind of structured work an agent can review at scale. An agentic reviewer can pull the relevant coding

policy, check eligibility, reason through whether a diagnosis justifies a procedure, and return a decision with a written rationale and a confidence score. Classical methods complement the agent rather than compete with it: unsupervised clustering and anomaly detection surface providers whose billing diverges from their peers without labeled training data, which is how new abuse patterns get found in the first place (De Meulemeester et al., 2025). Cotiviti has already staked out this ground, arguing that AI is a tool rather than a solution and that its value comes from pairing models with human expertise and prior results to learn from (Arnold & Periyanan, 2025). The point is not to replace reviewers. It is to give them a first pass that is faster, more consistent, and explained.

Threats

The main risk is the same property that makes these models useful: they produce fluent output that can be wrong. In a study of foundation models on medical tasks, 91.8% of surveyed clinicians had encountered a medical hallucination and 84.7% believed it could cause patient harm (Kim et al., 2025). A subtler risk is specific to using a model as the decision-maker. An agent that scores a claim is an instance of LLM-as-a-judge, and these systems are systematically overconfident: their stated confidence runs well ahead of how often they are actually right (Tian et al., 2025). In payment, an overconfident wrong answer is a denied claim that should have been paid, or an overpayment that clears review. Bias is a third risk, since a model trained on past decisions can repeat past mistakes. Auditability is the fourth, and it is where healthcare differs from most domains. A payer has to show why a decision was made, both to a regulator and to the provider appealing it. The same medical-hallucination study found chain-of-thought reasoning reduced hallucinations in most of its tests, partly because explicit reasoning steps let the system catch its own errors (Kim et al., 2025). Reasoning that is visible is also reasoning that can be checked.

Recommendation

Cotiviti should build agentic claim review as a human-in-the-loop system with a full audit trail, and treat that design as the differentiator rather than a compliance cost. It maps cleanly onto the company's own four principles: AI as a tool, AI used to improve results, AI governed responsibly for accuracy, transparency, bias, and accountability, and AI that supports rather than replaces human expertise (Arnold & Periyanan, 2025). Cotiviti already draws that last line in practice, using AI to prepare clinical content but not to make the decision, which is the human-in-the-loop posture this design assumes. The agent's chain-of-thought steps produce the rationale as a byproduct, so the reasoning is auditable by construction. RAG grounds each decision in the actual policy document, so a reviewer can trace it to source. Because the agent is overconfident by default, the confidence score should be calibrated and used as a routing threshold: clear the easy cases, and send the rest to people. Classical anomaly detection runs alongside as a second signal, so the system never leans on the language model alone. Operationally, each agent should carry an identity, a kill switch, and lifecycle logging, the controls that keep a fleet of agents accountable as it scales (Prakash et al., 2026). This is defense in depth with a person at the end. In payment integrity, a decision is only worth as much as the case behind it. The agent that wins is the one that can show its work.

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